Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A molecule M associates in a given solvent according to the equation $\mathrm{M} \rightleftharpoons (\mathrm{M})_n$. For a certain concentration of M, the van't Hoff factor was found to be 0.9 and the fraction of associated molecules was 0.2. The value of n is:

(A) 3

(B) 5

(C) 2

(D) 4

Q.2

A solution containing $0.85\ \mathrm{g}$ of $\mathrm{ZnCl_2}$ in $125.0\ \mathrm{g}$ of water freezes at $-0.23^{\circ}\mathrm{C}$. The apparent degree of dissociation of the salt is (K_f for water $= 1.86\ \mathrm{K\ kgmol^{-1}}$, atomic mass: $\mathrm{Zn} = 65.3$ and $\mathrm{Cl} = 35.5$)

(A) 1.36%

(B) 73.5%

(C) 7.35%

(D) 2.47%

Q.3

0.5 molal aqueous solution of a weak acid (HX) is 20% ionised. If K_f for water is $1.86\ \mathrm{K\ kg\ mol^{-1}}$, the lowering in freezing point of the solution is

(A) $0.56\ \mathrm{K}$

(B) $1.12\ \mathrm{K}$

(C) $-0.56\ \mathrm{K}$

(D) $-1.12\ \mathrm{K}$

Q.4

$200\ \mathrm{mL}$ of an aqueous solution of a protein contains its $1.26\ \mathrm{g}$. The osmotic pressure of this solution at $300\ \mathrm{K}$ is found to be 2.57×10^{-3} bar. The molar mass of protein will be
$\left(\mathrm{R} = 0.083\ \mathrm{Lbarmol^{-1}\ K^{-1}}\right)$

(A) $51022\ \mathrm{g\ mol^{-1}}$

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(B) $122044\ \text{g mol}^{-1}$

(C) $31011\ \text{g mol}^{-1}$

(D) $61038\ \text{g mol}^{-1}$

Q.5

Freezing point of an aqueous solution is $(-0.186)^{\circ}\text{C}$. Elevation of boiling point of the same solution is $K_b = 0.512^{\circ}\text{C}, K_f = 1.86^{\circ}\text{C}$, find the increase in
boiling point.

(A) 0.186°C

(B) 0.0512°C

(C) 0.092°C

(D) 0.2372°C

Q.6

The relationship between osmotic pressure at $273\ \text{K}$ when $10\ \text{g}$ glucose (P_1), $10\ \text{g}$ urea (P_2), and $10\ \text{g}$ sucrose (P_3) are dissolved in $250\ \text{mL}$ of water is

(A) $\text{P}_1 > \text{P}_2 > \text{P}_3$

(B) $\text{P}_3 > \text{P}_1 > \text{P}_2$

(C) $\text{P}_2 > \text{P}_1 > \text{P}_3$

(D) $\text{P}_2 > \text{P}_3 > \text{P}_1$

Q.7

Which observation(s) reflect(s) colligative
properties?
(i) A $0.5\ \text{m}$ NaBr solution has a higher vapour pressure than a $0.5\ \text{m}\text{BaCl}_2$ solution at the same temperature
(ii) Pure water freezes at the higher temperature than pure methanol
(iii) A $0.1\ \text{m}\text{NaOH}$ solution freezes at a lower temperature than pure water Choose the correct answer from the codes given below

(A) (i), (ii) and (iii)

(B) (i) and (ii)

(C) (ii) and (iii)

(D) (i) and (iii)

Q.8

The difference between the boiling point and freezing point of an aqueous solution containing sucrose (

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molecular $\mathrm{wt} = 342\ \mathrm{g\ mole^{-1}}$) in $100\ \mathrm{g}$ of water is $105^{\circ}\mathrm{C}$. If $\mathrm{K_f}$ and $\mathrm{K_b}$ of water are 1.86 and $0.51\ \mathrm{K\ kg\ mol^{-1}}$ respectively, the weight of sucrose in the solution is about

(A) $34.2\ \mathrm{g}$

(B) $342\ \mathrm{g}$

(C) $7.2\ \mathrm{g}$

(D) $72\ \mathrm{g}$

Q.9

A solution containing $1.8\ \mathrm{g}$ of a compound (empirical formula $\mathrm{CH_2O}$) in $40\ \mathrm{g}$ of water is observed to freeze at $-0.465^{\circ}\mathrm{C}$. The molecular formula of the compound is ($\mathrm{K_f}$ of water $= 1.86\ \mathrm{kg\ K\ mol^{-1}}$)

(A) $\mathrm{C_2H_4O_2}$

(B) $\mathrm{C_3H_6O_3}$

(C) $\mathrm{C_4H_8O_4}$

(D) $\mathrm{C_6H_{12}O_6}$

Q.10

Freezing point of an aqueous solution is $-0.186^{\circ}\mathrm{C}$. If the values of K_b and K_f of water are respectively $0.52\ \mathrm{K\ kg\ mol^{-1}}$ and $1.86\ \mathrm{K\ kg\ mol^{-1}}$, then the
elevation of boiling point of the solution in K is

(A) 0.52

(B) 1.04

(C) 1.34

(D) 0.052

Q.11

If the elevation in boiling point of a solution of $10\ \mathrm{g}$ of solute ($\mathrm{mol.\ wt.} = 100$) in $100\ \mathrm{g}$ of water is $\Delta\mathrm{T_b}$, the ebullioscopic constant of water is

(A) 10

(B) $10\Delta\mathrm{T_b}$

(C) $\Delta\mathrm{T_b}$

(D) $\frac{\Delta\mathrm{T_b}}{10}$

Q.12

Which one of the following statements is incorrect?

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(A) The correct order of osmotic pressure for $0.01\mathrm{M}$ aqueous solution of each compound is $\mathrm{BaCl_2} > \mathrm{KCl} > \mathrm{CH_3COOH} >$ Sucrose

(B) The osmotic pressure (π) of a solution is given by the equation $\pi = \mathrm{MRT}$, where M is the molarity of the solution

(C) Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction

(D) Two sucrose solutions of same molality prepared in different solvents will have the same freezing point depression

Q.13

π_1, π_2, π_3 and π_4 atm are the osmotic pressures of 5% (mass/volume) solutions of urea, fructose, sucrose and KCl respectively at certain temperature. The correct order of their magnitudes is :

(A) $\pi_1 > \pi_4 > \pi_2 > \pi_3$

(B) $\pi_1 > \pi_4 > \pi_2 > \pi_3$

(C) $\pi_4 > \pi_1 > \pi_2 > \pi_3$

(D) $\pi_4 > \pi_1 > \pi_3 > \pi_2$

Q.14

Which of the following has been arranged in the increasing order of freezing point?

(A) $0.025\mathrm{MKNO_3} < 0.1\mathrm{MNH_2CSNH_2} < 0.05\mathrm{MBaCl_2} < 0.1\mathrm{MNaCl}$

(B) $0.1\mathrm{MNaCl} < 0.05\mathrm{MBaCl_2} < 0.1\mathrm{MNH_2CSNH_2} < 0.025\mathrm{MKNO_3}$

(C) $0.1\mathrm{MNH_2CSNH_2} < 0.1\mathrm{MNaCl} < 0.05\mathrm{MBaCl_2} < 0.025\mathrm{MKNO_3}$

(D) $0.025\mathrm{MKNO_3} < 0.05\mathrm{MBaCl_2} < 0.1\mathrm{MNaCl} < 0.1\mathrm{MNH_2CSNH_2}$

Q.15

$0.010\mathrm{M}$ solution an acid HA freezes at $-0.0205^{\circ}\mathrm{C}$. If $\mathrm{K_f}$ for water is $1.860\ \mathrm{K\ kg\ mol^{-1}}$, the ionization
constant of the conjugate base of the acid will be (assume $0.010\mathrm{M} = 0.010\ \mathrm{m}$)

(A) 1.1×10^{-4}

(B) 1.1×10^{-3}

(C) 9.0×10^{-11}

(D) 9.0×10^{-12}

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Q.16

1.0 M aqueous solution of $\mathrm{CrCl_3 \cdot 6H_2O}$ freezes at $-5.58^{\circ}\mathrm{C}$. Assuming complete ionization of the hydrated complex, which of the following isomers conforms to the observation (K_f for water $= 1.86$ $\mathrm{Kkgmol^{-1}}$) ?

(A) $\left[\mathrm{Cr\left(H_2O\right)_3\ Cl_3}\right] .3\mathrm{H_2O}$

(B) $\left[\mathrm{Cr\left(H_2O\right)_4\ Cl_4}\right] \mathrm{Cl} .2\mathrm{H_2O}$

(C) $\left[\mathrm{Cr\left(H_2O\right)_5\ Cl}\right] \mathrm{Cl_2 \cdot H_2O}$

(D) $\left[\mathrm{Cr\left(H_2O\right)_6}\right] \mathrm{Cl_3}$

Q.17

A living cell contains a solution which is isotonic with $0.2\mathrm{M}$ glucose solution. What osmotic pressure develops when the cell is placed in $0.05\mathrm{MBaCl_2}$ solution at $300\ \mathrm{K}$?

(A) $1.23\ \mathrm{atm}$

(B) $3.69\ \mathrm{atm}$

(C) $6.15\ \mathrm{atm}$

(D) None of these

Q.18

At 760 torr pressure and $20^{\circ}\mathrm{C}$ temperature, 11 of water dissolves $0.04\mathrm{gm}$ of pure oxygen or $0.02\mathrm{gm}$ of pure nitrogen. Assuming that dry air is composed of 20% oxygen and 80% nitrogen (by volume), the masses (in $\mathrm{g/L}$) of oxygen and nitrogen dissolved by $1\ \mathrm{L}$ of water at $20^{\circ}\mathrm{C}$ exposed to air at a total pressure of 760 torr are
respectively:

(A) $0.008, 0.016$

(B) $0.016, 0.008$

(C) $0.16, 0.08$

(D) $0.04, 0.02$

Q.19

If sodium sulphate is considered to be completely dissociated into cations and anions in aqueous solution, the change in freezing point of water $\left(\Delta T_f\right)$, when $0.01\ \mathrm{mol}$ of sodium sulphate is dissolved in $1\ \mathrm{kg}$ of water, is $\left(K_f = 1.86\ \mathrm{K\ kg\ mol^{-1}}\right)$

(A) $0.372\ \mathrm{K}$

(B) $0.0558\ \mathrm{K}$

(C) $0.0744\ \mathrm{K}$

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(D) $0.0186\ \mathrm{K}$

Q.20

A solution of 0.5 g of a solute (molar mass $= 150$ g mol^{-1}) in 50 g of a solvent yields a boiling point elevation of 0.40 K. Another solution of 0.60 g of an unknown solute in the same mass of solvent exhibits a boiling point elevation of 0.8 K. The molar mass of unknown solute is :

(A) $60\ \mathrm{g\ mol^{-1}}$

(B) $90\ \mathrm{g\ mol^{-1}}$

(C) $120\ \mathrm{g\ mol^{-1}}$

(D) $180\ \mathrm{g\ mol^{-1}}$

Answers & Solutions

Q.1 Answer:

2

Solution:

van't Hoff factor (i) and the degree of association are related as below:

$i = 1 - \alpha\left(1 - \frac{1}{n}\right)$

$0.9 = 1 - 0.2\left(1 - \frac{1}{n}\right)$

On solving,

$\left(1 - \frac{1}{n}\right) = \frac{1}{2}$

$\frac{1}{n} = 1 - \frac{1}{2} = \frac{1}{2}$

$n = 2$

Q.2 Answer:

73.5%

Solution:

Mol.wt. $= \frac{k_f \times w \times 1000}{\Delta T_f \times W}$

$= \frac{1.86 \times 0.85 \times 1000}{0.23 \times 125} \approx 55\ \mathrm{g}$

Where $w = 0.85\ \mathrm{g}$

$W = 125\ \mathrm{g}$

$\Delta T_f = 0^\circ\mathrm{C} - (-23^\circ\mathrm{C}) = 23^\circ\mathrm{C}$

Now, $i = \frac{M_{\text{normal}}}{M_{\text{observed}}} = \frac{136.3}{55} = 2.47$

$\underset{1-\alpha}{\mathrm{ZnCl_2}} \leftrightharpoons \underset{\alpha}{\mathrm{Zn^{++}}} + \underset{2\alpha}{\mathrm{2Cl^-}}$

Van't Hoff factor (i)

$= \frac{1-\alpha+\alpha+2\alpha}{1} = 2.47$

- $\alpha = 0.735 = 73.5\%$

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Q.3 Answer:

$1.12\ \mathrm{K}$

Solution:

As $\Delta T_f = K_f \cdot m$

For,

	HX	$\rightleftharpoons$	$H^+ +$	X^-
$t = 0$	1	0	0	
$t = eq$	$(1 - 0.20)$	0.20	0.20	

Total no. of moles $= 1 - 0.20 + 0.20 + 0.20$

$= 1 + 0.20 = 1.2$

$\therefore \Delta T_f = 1.2 \times 1.86 \times 0.5 = 1.1160 \approx 1.12\ \mathrm{K}$

Q.4 Answer:

$61038\ \mathrm{g\ mol^{-1}}$

Solution:

Osmotic pressure $(\pi) = CRT$

$(\pi) = \frac{\text{wt} \times 1000}{\text{Molecular mass} \times V} RT$

$= 2.57 \times 10^{-3} = \frac{1.26 \times 1000}{\text{Mol.mass} \times 200} \times 0.083 \times 300$

Molecular mass $= 61038\ \mathrm{g}$

Q.5 Answer:

$0.0512^\circ \mathrm{C}$

Solution:

$\Delta T_b = K_b \frac{W_B}{M_B \times W_A} \times 1000$

$\Delta T_f = K_f \frac{W_B}{M_B \times W_A} \times 1000$

$\frac{\Delta T_b}{\Delta T_f} = \frac{K_b}{K_f} = \frac{\Delta T_b}{-0.186} = \frac{0.512}{1.86} = 0.0512^\circ \mathrm{C}.$

Q.6 Answer:

$P_2 > P_1 > P_3$

Solution:

Moles of glucose $\frac{10}{180} = 0.05,$

urea $= \frac{10}{60} = 0.16,$ **Sucrose** $= \frac{10}{342} = 0.029$

Hence osmotic presure $P_2 > P_1 > P_3 (\pi \propto C)$

Q.7 Answer:

(i) and (iii)

Solution:

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Colligative properties depends upon the no. of particles. Since methanol is non electrolyte hence cannot be considered.

Q.8 Answer:

72 g

Solution:

$(100+\Delta T_b)-(0-\Delta T_f)=105$

$\Delta T_b+\Delta T_f=5$

$m\left(K_b+K_f\right)=5$

$m=\frac{5}{2.37}$ **i.e.,** $\frac{5}{2.37}$ **moles in** 1000 g **water**

(or) $\frac{5}{2.37\times10}$ **moles in** 100 g **water**

$\therefore$ **Wt. of sucrose** $=\frac{5}{2.37\times10}\times342=72$ g

Q.9 Answer:

$C_6H_{12}O_6$

Solution:

$\Delta T_f=K_f\times m$

$M=\frac{1000\times K_f\times w_2(\text{ solute })}{\Delta T_f\times w_1(\text{ solvent })}$

$=\frac{1000\times1.86\times1.8}{0.465\times40}\Rightarrow M=180$

Molecular formula $=(\text{ empirical formula })_n$ $n=\frac{\text{Molecular mass}}{\text{Empirical formula mass}}=\frac{180}{30}=6$

Molecular formula $=(CH_2O)_6=C_6H_{12}O_6$

Q.10 Answer:

0.052

Solution:

$\Delta T_f=i\cdot K_f\cdot m;\Delta T_b=i.K_b\cdot m$

$\frac{\Delta T_f}{\Delta T_b}=\frac{K_f}{K_b}$

$\Delta T_f=0-(-0.186^\circ C)=0.186^\circ C$

$\frac{0.186}{\Delta T_b}=\frac{1.86}{0.52}\Rightarrow\Delta T_b=\frac{0.52\times0.186}{1.86}=0.052$

Q.11 Answer:

ΔT_b

Solution:

$$\Delta T_b=\frac{K_b\times w\times1000}{M\times W}$$

$$\therefore K_b=\frac{\Delta T_b\times100\times100}{10\times1000}=\Delta T_b$$

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Q.12 Answer:

Two sucrose solutions of same molality prepared in different solvents will have the same freezing point depression

Solution:

($\Delta T_f = K_f \times m \times i$. **Since** K_f **has different values for different solvents, hence even if** m **is the same** ΔT_f **will be different**

Q.13 Answer:

$\pi_4 > \pi_1 > \pi_2 > \pi_3$

Solution:

Moles of urea $= \frac{}{60}$;
moles of fructose $= \frac{5}{180}$
$C_{urea} = \frac{0.08 \times 1000}{100} \Rightarrow 0.833$
moles of sucrose $= \frac{5}{342}$
$C_{sucrose} = 0.0146 \times \frac{1000}{100} = 0.146$
moles of KCl **(effective)**
$= 2 \times \frac{5}{74.5} = \frac{5}{37.25}; \frac{0.134 \times 1000}{100} = 1.34$
$\pi \propto C$
this order is $\pi_4 > \pi_1 > \pi_2 > \pi_3$

Q.14 Answer:

$0.1MNaCl < 0.05MBaCl_2$
$< 0.1MNH_2CSNH_2 < 0.025MKNO_3$

Solution:

Greater is the effective molarity $(1 \times C)$**, higher the** ΔT_f **value and lower the freezing point.**
(i) $0.1MNaCl = i \times C = 2 \times 0.1 = 0.2$
(ii) $0.05MBaCl_2 = i \times C = 3 \times 0.05 = 0.15$
(iii) $0.1MNH_2CSNH_2 = i \times C = 1 \times 0.1 = 0.1$
(iv) $0.025MKNO_3 = i \times C = 2 \times 0.025 = 0.50$
Thus, order is $(i) < (ii) < (iii) < (iv)$

Q.15 Answer:

9.0×10^{-11}

Solution:

ΔT_f **(normal)** $= K_f m = 1.86 \times 0.01 = 0.0186;$
$i = \frac{\Delta T_{f(obs)}}{\Delta T_{f(nor)}} = \frac{0.0205}{0.0186} = 1.10 = 1 + \alpha$
$\alpha = 0.1$

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$\mathrm{K_a} = \frac{\mathrm{C}\alpha^2}{1-\alpha} = \frac{0.01 \times 0.1^2}{1-0.1} = \frac{1}{9} \times 10^{-3}$
$\mathrm{K_b} = \frac{\mathrm{K_w}}{\mathrm{K_a}} = 1.0 \times 10^{-14} \times 9 \times 10^3 = 9 \times 10^{-11}$

Q.16 Answer:

$[\mathrm{Cr}\,(\mathrm{H_2O})_5\,\mathrm{Cl}]\,\mathrm{Cl_2} \cdot \mathrm{H_2O}$

Solution:

$\mathrm{i} = \frac{5.58}{1.86} = 3;$
$[\mathrm{Cr}\,(\mathrm{H_2O})_5\,\mathrm{Cl}]\,\mathrm{Cl_2} \cdot \mathrm{H_2O} \longrightarrow$
$[\mathrm{Cr}\,(\mathrm{H_2O})_5\,\mathrm{Cl}]^{2+} + 2\mathrm{Cl}^-$

Q.17 Answer:

$1.23\ \mathrm{atm}$

Solution:

For non electrolyte solute (glucose) $i = 1$
For $\mathrm{BaCl_2} \rightleftharpoons \mathrm{Ba}^{2+} + 2\mathrm{Cl}^-$
$\mathrm{i} = 3(100\%$ **ionized** $)$
$\mathrm{i} = ($ **number of ions afterionization** $\times \alpha) + (1 - \alpha)$
$= (3 \times 1) + (1 - 1) = 3$
$\pi = (\mathrm{iC_1} - \mathrm{i_2C_2})\,\mathrm{RT}$
$\Rightarrow (1 \times 0.2 - 3 \times 0.05) 0.0821 \times 300$
$\Rightarrow 1.23\ \mathrm{atm}$

Q.18 Answer:

$0.008, 0.016$

Solution:

Mass of oxygen $= \frac{20}{100} \times 0.04 = 0.008\ \mathrm{g/L}$
Mass of nitrogen
$= \frac{80}{100} \times 0.02 = 0.016\ \mathrm{g/L}$

Q.19 Answer:

$0.0558\ \mathrm{K}$

Solution:

Sodium sulphate dissociates as
$\mathrm{Na_2SO_4(\,s)} \longrightarrow 2\mathrm{Na}^+ + \mathrm{SO_4^{--}}$
hence van't hoff factor $i = 3$
Now $\Delta T_f = ik_f \cdot m$
$= 3 \times 1.86 \times 0.01 = 0.0558\ \mathrm{K}$

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Q.20 Answer:

90 g mol^{-1}

Solution:

$$\frac{\Delta T_{b_1}}{\Delta T_{b_2}} \propto \frac{m_1}{m_2} \propto \frac{(\text{moles})_1}{(\text{moles})_2}$$ ∵ mass of solvent is same

$$\frac{0.40}{0.80} = \frac{0.5 \times x}{150 \times 0.60}$$

$$0.4x = 36$$

$$x = 90$$

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